

RESEARCH ARTICLE

Neural Network-Based Feedback Linearization Slip Control for Anti-Lock Braking Systems: A NARMA-L2 Approach

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Abstract

Anti-lock braking systems (ABS) are critical safety components in modern vehicles, requiring precise slip ratio control to maximize braking efficiency while maintaining vehicle stability. This paper presents a neural network-based feedback linearization approach using NARMA-L2 controller for ABS slip regulation. Through systematic training on synthetic ABS data generated from a quarter-car model, the NARMA-L2 controller was developed and compared against conventional PID control strategies. The study reveals important insights into the limitations of synthetic training data generated with idealized controllers and identifies key areas for future research in neural network-based ABS control.

Keywords: anti-lock braking system, NARMA-L2 controller, neural networks, feedback linearization, slip control, automotive safety

1. Introduction

Modern automotive safety systems rely heavily on precise control of vehicle dynamics during critical braking scenarios. Anti-lock braking systems (ABS) represent one of the most significant advances in automotive safety technology, preventing wheel lockup and maintaining vehicle steerability during emergency braking situations. The fundamental challenge in ABS design lies in maintaining the wheel slip ratio at an optimal value that maximizes the friction coefficient between tire and road surface.

Traditional ABS controllers, primarily based on PID control strategies, have proven effective in many scenarios but suffer from limitations when dealing with the inherent nonlinearities of tire-road friction characteristics and varying road conditions. The tire-road friction curve exhibits a complex nonlinear relationship with slip ratio, characterized by a peak friction coefficient at an optimal slip ratio (typically around 15 % to 25 %), beyond which the friction coefficient decreases rapidly, leading to wheel lockup.

Neural network-based control approaches have emerged as promising alternatives for handling such nonlinear systems. Among these, the NARMA-L2 control strategy offers a systematic approach to feedback linearization of nonlinear systems. This method transforms the nonlinear system dynamics into a linear relationship through the cancellation of system nonlinearities, enabling the application of linear control techniques to inherently nonlinear systems.

The state-of-the-art in ABS control continues to rely predominantly on classical control approaches, with limited exploration of advanced neural network techniques in real-world applications. Existing neural network implementations often lack comprehensive comparison with established PID controllers and fail to address the practical challenges of training data generation and controller robustness. Moreover, the influence of training data quality on neural network controller

performance remains an understudied aspect that significantly impacts the practical viability of such approaches.

To address these gaps, this paper presents a comprehensive implementation and evaluation of NARMA-L2 controllers for ABS slip regulation. We developed a complete simulation framework incorporating a quarter-car ABS model, implemented both f-network and g-network components of the NARMA-L2 architecture, and conducted detailed comparative analysis against conventional PID controllers. The study reveals both the potential benefits and current limitations of our neural network-based ABS control, providing insights for future research directions.

2. Mathematical Modeling and System Description

2.1 Quarter-Car ABS Model

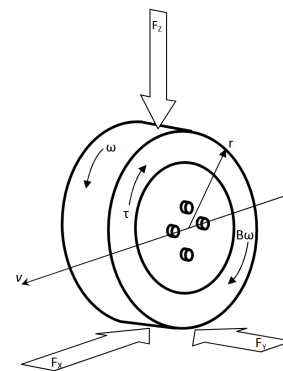


Figure 1. Quarter car braking model

The ABS system dynamics are modeled using a quarter-car approach, which captures the essential physics of wheel-vehicle interaction during braking. The mathematical model consists of three coupled differential equations describing vehicle dy-

namics, wheel rotational dynamics, and electro-mechanical brake system behavior.

The vehicle longitudinal dynamics are governed by:

$$M \cdot \dot{v} = -\mu(\lambda)F_z - C_x \cdot v^2 \quad (1)$$

where M represents the vehicle mass, v is the vehicle velocity, $\mu(\lambda)$ is the slip-dependent friction coefficient, F_z is the normal force, and C_x is the aerodynamic drag coefficient.

The wheel rotational dynamics are described by:

$$I \cdot \dot{\omega} = \mu(\lambda)F_z \cdot r - B \cdot \omega - \tau_b \quad (2)$$

where I is the wheel moment of inertia, ω is the wheel angular velocity, r is the wheel radius, B is the rolling friction coefficient, and τ_b is the applied braking torque.

The electro-mechanical brake system dynamics are modeled as:

$$\dot{\tau}_b = \frac{-\tau_b + K_b \cdot u}{\tau} \quad (3)$$

where K_b is the braking gain, u is the control input (brake pressure command), and τ is the system time constant.

The slip ratio, which serves as the controlled variable, is defined as:

$$\lambda = \frac{v - r \cdot \omega}{v} \quad (4)$$

2.2 Tire-Road Friction Model

The tire-road friction characteristic is modeled using the following relationship:

$$\mu(\lambda) = \frac{2\mu_0\lambda_0|\lambda|}{\lambda_0^2 + \lambda^2} \quad (5)$$

where μ_0 is the maximum friction coefficient and λ_0 is the optimal slip ratio. This model captures the characteristic peak in friction coefficient at the optimal slip ratio, followed by a gradual decrease as slip increases.

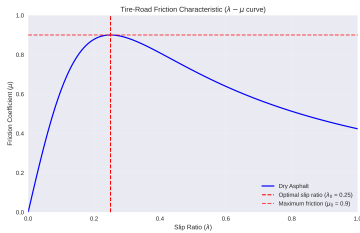


Figure 2. Tire-road friction characteristic showing the relationship between slip ratio and friction coefficient. The optimal slip ratio $\lambda_0 = 0.25$ corresponds to maximum friction coefficient $\mu_0 = 0.9$.

2.3 State Equations

From Eqs. (1)–(5), we obtain the nonlinear state equations:

$$\begin{aligned} \dot{\omega} &= \frac{\mu(\lambda)F_z r}{I} - \frac{B\omega}{I} - \frac{\tau_b}{I} \\ \dot{v} &= -\frac{\mu(\lambda)F_z}{M} - \frac{C_x v^2}{M} \\ \dot{\tau}_b &= \frac{-\tau_b + K_b v_b}{\tau} \end{aligned} \quad (6)$$

The state variables are defined as $x_1 = \omega$, $x_2 = v$, and $x_3 = \tau_b$. The control input is $u = v_b$, and the controlled output is $y = \lambda$.

Thus, the system model can be written in state-space form as:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, u), \quad y = \mathbf{g}(\mathbf{x}) \quad (7)$$

3. NARMA-L2 Controller Design

3.1 Theoretical Foundation

The NARMA-L2 controller is based on the feedback linearization principle, where the nonlinear system dynamics are transformed into a linear relationship. The key equation governing the NARMA-L2 model is:

$$y(k+d) = f(\cdot) + g(\cdot) \cdot u(k+1) \quad (8)$$

where $y(k+d)$ represents the future system output (slip ratio), $f(\cdot)$ is a nonlinear function depending on past outputs and inputs, $g(\cdot)$ approximates the system sensitivity to control input, and d is the system delay.

The functions $f(\cdot)$ and $g(\cdot)$ are approximated using separate neural networks trained on system data. The f-network captures the autonomous system behavior, while the g-network models the control sensitivity.

3.2 Neural Network Architecture

We chose $d = 2$ as the delay for this paper. Both f-network and g-network employ a single hidden layer architecture with five neurons, using hyperbolic tangent activation functions in the hidden layer and linear activation in the output layer. The input vector for both networks consists of:

$$\mathbf{x}(k) = [y(k), y(k-1), y(k-2), u(k), u(k-1), u(k-2)]^T \quad (9)$$

This structure captures the system's dependence on both past outputs and control inputs, consistent with the NARMA model formulation.

3.3 Training Strategy

The NARMA-L2 training follows a systematic two-step approach:

Step 1: G-network Training The g-network is trained first to approximate the system's control sensitivity. Initial g-values are estimated and used to establish the baseline control sensitivity characteristics. Here we state that $g_k = o(f_k)$ in order to have some values to predict and so

$$g_k = (y[k+1] - y[k]) / (u[k+1] - u[k]) \quad (10)$$

Step 2: F-network Training Following g-network training, corrected targets for the f-network are calculated using:

$$f_k = y(k+1) - g_k \cdot u(k+1) \quad (11)$$

This approach ensures that the f-network learns the autonomous system dynamics after accounting for control influence.

3.4 Control Law Implementation

The NARMA-L2 control law is implemented by solving for the required control input:

$$u(k+1) = \frac{\gamma_{ref} - f_k}{g_k} \quad (12)$$

where γ_{ref} is the reference slip ratio ($\lambda_0 = 0.25$), and f_k and g_k are the neural network predictions.

4. Training Data Generation and Methodology

4.1 Synthetic Data Generation

Table 1. Model and Controller Parameters

Symbol	Description	Value	Unit
<i>Vehicle and Environment Parameters</i>			
M	Vehicle mass	440	kg
I	Wheel moment of inertia	1.6	kg·m ²
r	Wheel radius	0.3	m
B	Rolling resistance coefficient	0.08	kg·m ² /s
C_x	Aerodynamic drag coefficient	0.856	kg/m
K_b	Braking gain	0.8	–
τ	Actuator time constant	0.003	s
g	Gravitational acceleration	9.81	m/s ²
F_z	Normal force ($M \cdot g$)	4316.4	N
μ_0	Max friction coefficient	0.9	–
λ_0	Optimal slip ratio	0.25	–
<i>PID Controller Parameters</i>			
K_P	Proportional gain	3000	–
T_I	Integral time constant	0.1	s
T_D	Derivative time constant	0.01	s
α	Derivative lag factor	0.1	–

Training data for the neural networks were generated using the quarter-car ABS model under a set of randomized braking scenarios. A total of 50 distinct simulations were conducted, each lasting 4 seconds with a sampling time of 0.01 seconds, resulting in approximately 20,000 data points.

To introduce variability and ensure generalization, the following parameters were randomized at the beginning of each scenario:

- Initial vehicle speed v_0 was sampled uniformly from 20 to 40 m/s.
- Initial slip ratio λ_0 was indirectly defined by drawing a slip value between 0.1 and 0.4.
- Initial wheel angular velocity ω_0 was computed from v_0 and the slip ratio to maintain physical consistency.
- Initial brake torque $\tau_b(0)$ was sampled uniformly from 0 to 20 Nm.

The data generation process employed a PID controller to regulate slip ratio, creating realistic control scenarios for neural network training. However, this approach introduced an important limitation: the training data reflected idealized

PID control behavior, potentially biasing the neural network toward replicating rather than improving upon conventional control strategies.

The structure of the PID controller is given by:

$$U(s) = K_P \cdot \left(\frac{1 + T_D s}{1 + \alpha T_D s} \cdot \frac{1 + T_I s}{T_I s} \right) \cdot E(s) \quad (13)$$

where:

- $E(s) = Y_{ref}(s) - Y(s)$ is the error signal between the reference input $Y_{ref}(s)$ and the actual system output $Y(s)$,
- $U(s)$ is the plant input signal (control effort),
- K_P is the proportional gain,
- T_D is the derivative time constant,
- T_I is the integral time constant,
- α is the lag factor in the derivative component.

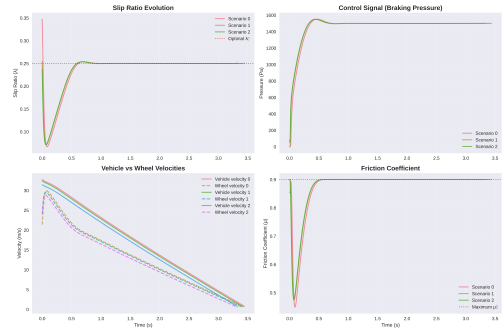


Figure 3. Visualization of 3 different scenarios and the evolution of their characteristics : (a) slip ratio, (b) control input, (c) vehicle and wheel velocity and (d) friction.

4.2 Data Preprocessing

Input features were standardized using StandardScaler to ensure proper neural network training convergence. The dataset was split into training (80%) and testing (20%) sets, with temporal ordering preserved to maintain the sequential nature of the control problem.

Data quality analysis revealed well-distributed slip ratios around the optimal value, with control inputs spanning the full operational range of the brake system. Vehicle velocities ranged from 40 m/s to near-zero, covering typical braking scenarios from highway speeds to complete stops.

5. Experimental Results and Analysis

5.1 Training Performance

The neural networks were respectively trained for 5 and 10 epochs using the Adam optimizer with a learning rate of 0.001. We didn't trained the neural networks longer because overfitting was clearly showing in the results, even though the learning curves were normal. Training convergence was achieved for both f-network and g-network, with MSE (mean squared error) losses reaching acceptable levels.

The g-network demonstrated stable convergence with final training loss of approximately 10^{-4} , which is logical due

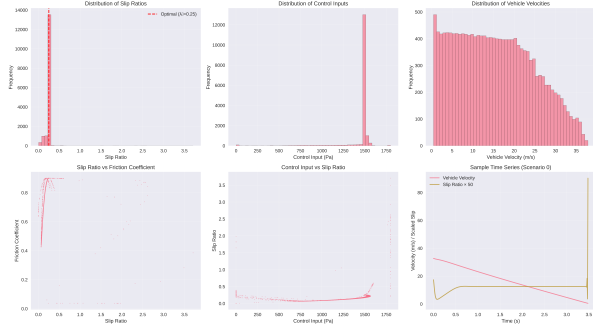


Figure 4. Distribution of key variables in the training dataset: (a) slip ratio distribution, (b) control input distribution, (c) vehicle velocity distribution, (d) slip-friction relationship and (e) slip-input relationship.

to the way we defined it. The f-network training showed more variability but achieved convergence with final loss of approximately 10^{-1} .

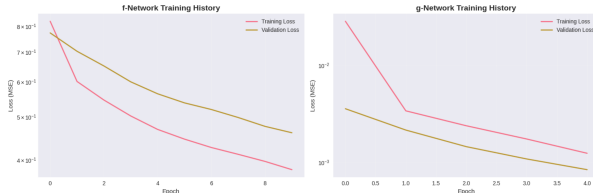


Figure 5. Training history for both neural networks: (a) g-network loss evolution, (b) f-network loss evolution showing convergence characteristics.

5.2 Prediction Analysis

We analyzed the NARMA-L2 controller's predictions on a defined scenario and it revealed some key points:

Slip Ratio Tracking: The NARMA-L2 controller predicted almost the same values of slip ratio compared to the PID controller even though it some deviations.

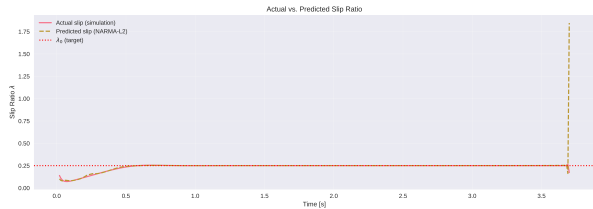


Figure 6. Slip Ratio prediction.

Brake Torque: Here, the NARMA-L2 controller shows some inconsistencies about the predicted values and we can see some spikes in the curve. Moreover, the predicted values were almost always higher than the required ones according to the PID controller.

5.3 Comparative Performance Analysis

Then, the NARMA-L2 controller was compared against a conventional PID controller in terms of slip ratio tracking performance and control effort. The comparison revealed several key findings:

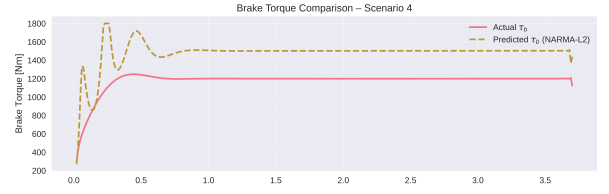


Figure 7. Brake Torque prediction.

Slip Ratio Tracking: The NARMA-L2 controller showed way less precise slip ratio tracking compared to the PID controller. While the PID controller maintained slip ratio close to the optimal value of 0.25, the NARMA-L2 controller exhibited larger deviations, less stable regulation and seems to reach a lower slip ratio value than expected.

Control Effort: Notably, the NARMA-L2 controller produced spikes and almost consistently applied lower brake pressures compared to the PID controller. This characteristic, while resulting in suboptimal slip tracking, suggests potential benefits for brake system longevity and reducing energy consumption.

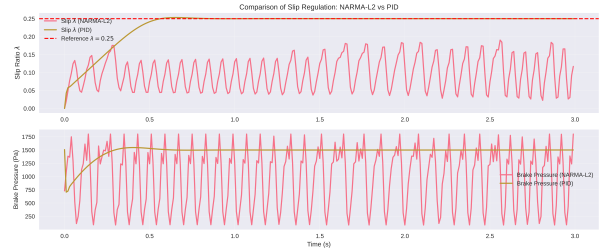


Figure 8. Comparative performance analysis: (a) slip ratio tracking comparison and (b) brake pressure comparison.

System Response: Both controllers achieved similar vehicle deceleration profiles but the overall braking effectiveness remained higher and more stable for the PID controller.

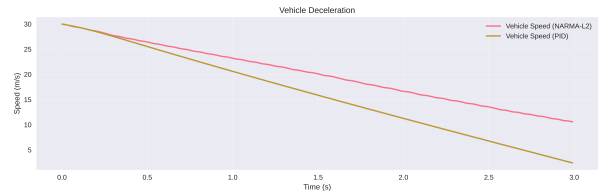


Figure 9. Comparative performance analysis: vehicle deceleration profiles for both NARMA-L2 and PID controllers.

6. Discussion

6.1 Performance Limitations and Insights

The experimental results reveal important insights into the practical challenges of implementing neural network-based ABS controllers. The primary limitation observed was the NARMA-L2 controller's inability to achieve precise slip ratio tracking compared to the conventional PID approach.

This performance gap can be attributed to several factors. First, the training data generation process using an idealized

PID controller may have created a biased dataset that did not adequately represent the full range of system behaviors required for robust neural network training. The "too perfect" nature of the PID-generated training data likely limited the neural network's ability to learn optimal control strategies that could surpass conventional approaches.

Second, the NARMA-L2 model's assumption of system linearity after feedback linearization may not fully capture the complex, time-varying nature of tire-road friction dynamics, particularly during transient braking conditions where road surface characteristics can change rapidly.

6.2 Unexpected Benefits and Future Opportunities

Despite the tracking performance limitations, the NARMA-L2 controller demonstrated a promising characteristic: consistently lower brake pressure application while achieving comparable overall braking performance. This observation opens several avenues for future research and development.

The reduced brake pressure requirement could translate to significant practical benefits, including extended brake system component life, reduced brake fade during repeated applications, and improved energy efficiency in electro-mechanical brake systems. In the context of electric and hybrid vehicles, where regenerative braking integration is crucial, the NARMA-L2 approach's gentler brake application profile could facilitate smoother transitions between regenerative and friction braking.

6.3 Training Data Quality Considerations

A critical insight from this study concerns the importance of training data quality in neural network-based control applications. The use of PID-generated training data created an inherent bias that limited the neural network's learning potential. Future implementations should consider alternative data generation strategies, such as:

- Multi-controller training data incorporating various control strategies
- Real-world driving data collection under diverse braking scenarios
- Adversarial training approaches to improve controller robustness
- Incorporation of system uncertainties and disturbances in training data

6.4 Implications for ABS System Design

The findings suggest that neural network-based ABS controllers may be most beneficial in applications where the control objectives extend beyond simple slip ratio tracking. For instance, in autonomous vehicle applications where multiple objectives must be balanced (safety, comfort, energy efficiency, component longevity), the NARMA-L2 approach's multi-faceted benefits could prove valuable.

Additionally, the neural network framework provides flexibility to incorporate additional sensor inputs and environmental conditions, potentially enabling more adaptive and context-aware braking control strategies.

7. Conclusion

This study demonstrated that while the NARMA-L2 neural network-based ABS controller did not outperform a conventional PID controller in slip ratio tracking, however it still offered a reduced brake pressure for comparable deceleration. These results suggest that such controllers may hold promise in applications prioritizing comfort, component longevity, or energy efficiency.

However, several limitations must be acknowledged. The use of a quarter-car simulation, although appropriate for initial validation, does not fully capture real-world dynamics such as tire wear, road surface variation, and vehicle body interactions. Moreover, the NARMA-L2 controller's neural network architecture was relatively simple, and the training data—derived solely from an idealized PID controller—introduced bias that likely constrained learning.

To build on these findings, future work should focus on:

- **Hybrid control architectures** combining the precision of PID with the adaptability of neural networks;
- **Multi-objective training frameworks** that explicitly balance slip tracking, comfort, energy usage, and component wear;
- **Robust, diverse training datasets** derived from real driving scenarios, including noise and disturbances;
- **Online learning mechanisms** for real-time adaptation to changing road and vehicle conditions;
- **Real-world validation** on full-scale vehicles to confirm simulation results and expose practical deployment challenges.

Ultimately, this work try to lay a foundation for the development of intelligent braking systems that go beyond traditional control goals. Neural network-based ABS controllers, particularly when combined with adaptive or hybrid strategies, represent a promising path toward safer, more comfortable, and efficient braking in next-generation vehicles.